



QUANTITATIVE METHODS 1 · CHAPTER 4

Bivariate Distributions

Exercise series

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1. Reading a relationship

A scatter plot of advertising expenditure and sales slopes upward, with points close to a line except for one distant observation.

- 1 Describe direction, form and strength.
- 2 What should be investigated about the distant point?
- 3 Does the plot prove advertising causes sales?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

2. Covariance and correlation

For $x=(1,2,3,4)$ and $y=(2,3,5,6)$ calculate population covariance and correlation.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

3. Least-squares regression

Use the data in Exercise 2 to estimate $y=a+bx$. Forecast y for $x=3.5$ and calculate the residual at (3,5).

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

4. R^2 , interpolation and extrapolation

Suppose a model fitted for advertising budgets from EUR 1,000 to EUR 10,000 has $R^2 = .81$. Interpret R^2 . Classify predictions at EUR 7,000 and EUR 20,000 and explain the risk.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

5. From association to a decision

A retailer finds $r=.85$ between discounts and weekly sales and recommends permanent large discounts. Give a statistically responsible response including at least four checks.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

6. Complete regression from paired data

For six stores, weekly advertising x (€000) and sales y (€000) are:

x	1	2	3	4	5	6
y	3	5	4	7	8	9

- 1 Draw and describe the scatter plot.
- 2 Given $\bar{x} = 3.5$, $\bar{y} = 6$, $\Sigma(x - \bar{x})^2 = 17.5$ and $\Sigma(x - \bar{x})(y - \bar{y}) = 21$, estimate the regression line.
- 3 Predict sales for $x = 4.5$.
- 4 Calculate the residual for the store with $x = 4$.

Method prompt. Use $b = S_{xy}/S_{xx}$ and $a = \bar{y} - b\bar{x}$. A residual is observed minus fitted.

7. Regression diagnosis and responsible decision

A model linking store traffic to temperature reports $r = 0.78$ and $R^2 = 0.61$. Its residual plot is U-shaped, one summer weekend has a very large residual, and the manager wants to forecast a temperature 12°C above the observed maximum.

- 1 Interpret r and R^2 correctly.
- 2 Diagnose the residual pattern.
- 3 Discuss the unusual weekend and the proposed forecast.
- 4 Recommend the next analytical steps.

Method prompt. Separate strength, functional form, unusual observations, extrapolation and causality.